

3.1.1 Programming Language Types

پروگرامنگ زبانوں کی تین اقسام ہیں: لو لیول (Low Level) جو ہائری 0 اور 1 میں ہوتی ہے، مڈ لیول (Mid Level) یعنی...

CS XI

Chapter 3: Programming Fundamentals

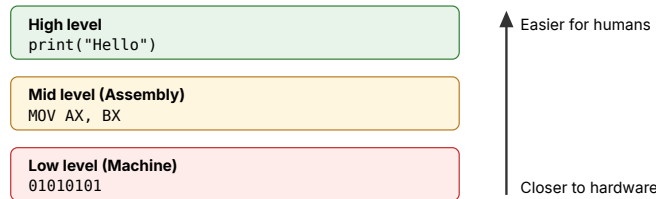
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3.1.1 Programming Language Types

LEARN IT

Programming languages are classified into **three main types**, depending on how close they are to the computer's own language (binary 0s and 1s) or to human language.

Type	Description	Examples
Low level	Closest to the native language of the computer (binary 0s and 1s). Very difficult for humans to read and write.	Machine language (binary)
Mid level	Bridges the gap. It can access hardware but is easier to read than binary because it uses short words called mnemonics (e.g. MOV, ADD).	Assembly language
High level	Designed to be easy for humans to read and write. Uses English-like words.	C++, Java, Python



The three levels of programming languages, with a sample line from each.

Common high-level languages

Language	Key idea	Used for
C++	Powerful and closer to hardware.	System software, operating systems, high-performance computer games.
Java	"Write once, run anywhere."	Large enterprise applications, Android mobile apps, server programming.
Python	Simple, readable syntax.	Data Science, Artificial Intelligence, Web Development, and learning to program.

WORKED EXAMPLE

Same task, three levels. Suppose you want to copy a value.

- Machine language: a pattern like `01010101` — you would need a chart to know what it means.
- Assembly: `MOV AX, BX` — "move the value in register BX into register AX". Shorter words, but you still need to know the hardware.
- Python: `a = b` — you simply say "a gets the value of b".

The higher the level, the less you need to know about the hardware and the faster you can write the program.

REMEMBER IT

"Low = Like the machine, High = Human." Mid level sits in the Middle and uses Mnemonics (both start with M).

COMMON MISTAKE

"Low level" does not mean "bad" or "weak". Low-level code runs very fast and controls hardware directly. It is just harder for humans to write.

اردو میں سمجھیں

پروگرامنگ زبانوں کی تین اقسام ہیں: لو لیول (Low Level) جو بائری 0 اور 1 میں ہوتی ہے، مڈ لیول (Mid Level) یعنی اسمبلی جو مختصر الفاظ (Mnemonics) استعمال کرتی ہے، اور ہائی لیول (High Level) جیسے پائتھن، جاوا اور سی پلس پلس جو انگریزی جیسے الفاظ استعمال کرتی ہیں اور انسانوں کے لیے آسان ہیں۔

CHECK YOURSELF

- Which level is `MOV AX, BX`?
- Which high-level language is famous for "write once, run anywhere"?
- Give one use of C++.

Answers: 1. Mid level (assembly language). 2. Java. 3. Any one: system software, operating systems, high-performance games.

Next lecture: 3.2 Career Growth with Python